

Regulatory Clearance for SaMD

[Software as Medical Device]

June 5th, 10:00-10:30

Chao Wen-Cheng. MD, PhD.

Department of Critical Care Medicine, Taichung Veterans General Hospital, Taiwan.



Outline

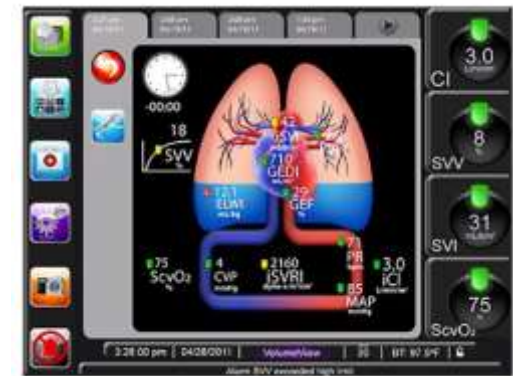
- **Unmet need and opportunity of AI models in critical care medicine**
- **3P of AI model: Paper-Patent-Product (SaMD)**
- **Collaboration for external validation and quality management system (QMS) approval of SaMD**

SaMD: Software as Medical Device

Challenges and unmet needs of critical care

● Complex disease, multi-organ involvement, data in distinct devices

● Burnout of critical care professionals, time to integrate data



- Critical care clinicians have some of the highest rates of burnout of all specialties—as high as 70% prevalence among physicians, and up to 80% among nurses.

*Mara Buchbinder, Tania Jenkins.
Ann Am Thorac Soc. 2022 Sep;19(9):1443-1445.*

Burnout in ICU: Challenge and Opportunity

Physicians, as future users of medical AI, should be actively involved in designing AI-aided critical care systems to ease the workload in critical care.

1. Establishment the critical care database based on global standard
2. Explainable AI to convince the physicians and nurses
3. Lading of actionable AI that align with workflow in the ICU

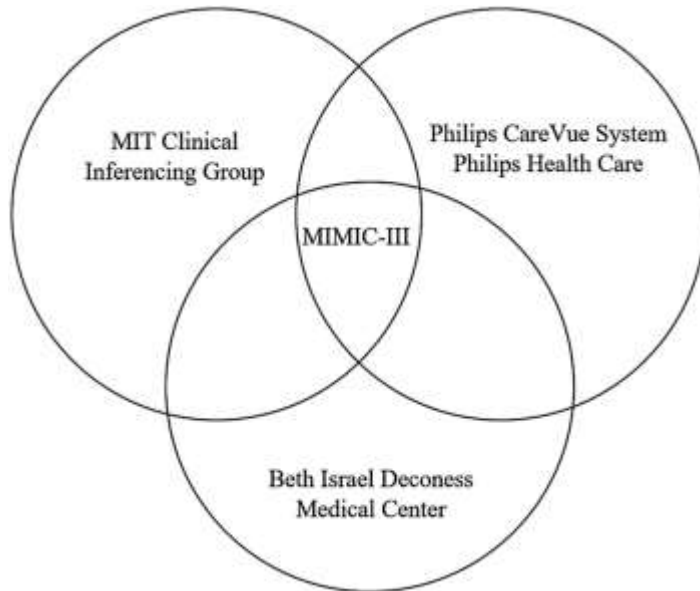
Critical Care Databases

MIT

Massachusetts Institute of
Technology

PHILIPS

Medical **I**nformation **M**art
for **I**ntensive **C**are (MIMIC)



Beth Israel Deaconess
Medical Center



HARVARD MEDICAL SCHOOL
TEACHING HOSPITAL

Items	eICU-CRD	MIMIC-III
Country	United States	United States
Data	Multicenter	Single-center
Year	2014–2015	2001–2012
Number of units	335	1
Number of hospitals	208	1
Number of patients	139,367	38,597
Number of admissions	200,859	53,423

**Standard of database
[Inter-operability]**

MIMIC IV: 2008-2019

FiO1: 40% vs .40

FHIR (Fast Healthcare Interoperability Resources)

TCVGH Critical Care Databank

7_{yr}
2015~2021

24_{types}
Dx, Lab &
Image

339
features

72m
data

33508
ICU admission

01 Personal information

02 ICU severity scores

03 Discharge diagnosis

04 Operation

05 Post history

06 Laboratory examination

08 Arterial blood gas

09 Medication

10 Blood transfusion

11 Renal replacement therapy

12 Vital signs in ICU

13 I/O and nutrition

14 Microbiological exam

15 Mechanical ventilation

16 Cause of death data

17 DNR order

18 Catheter use in ICU

19 Chest X ray

20 Past laboratory data

21 Radiological exam

22 Medication dosage at ICU

23 Medical history of chronic renal disease

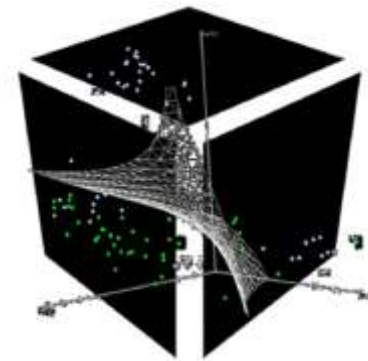
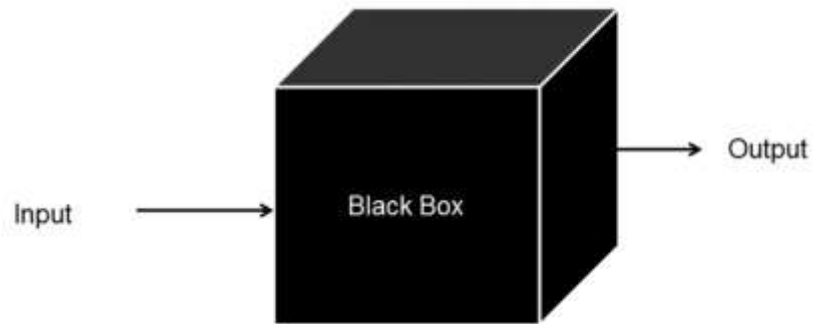
24 Dialysis record

Taiwan's National Health Insurance Research Database

Ethics guidelines for trustworthy AI

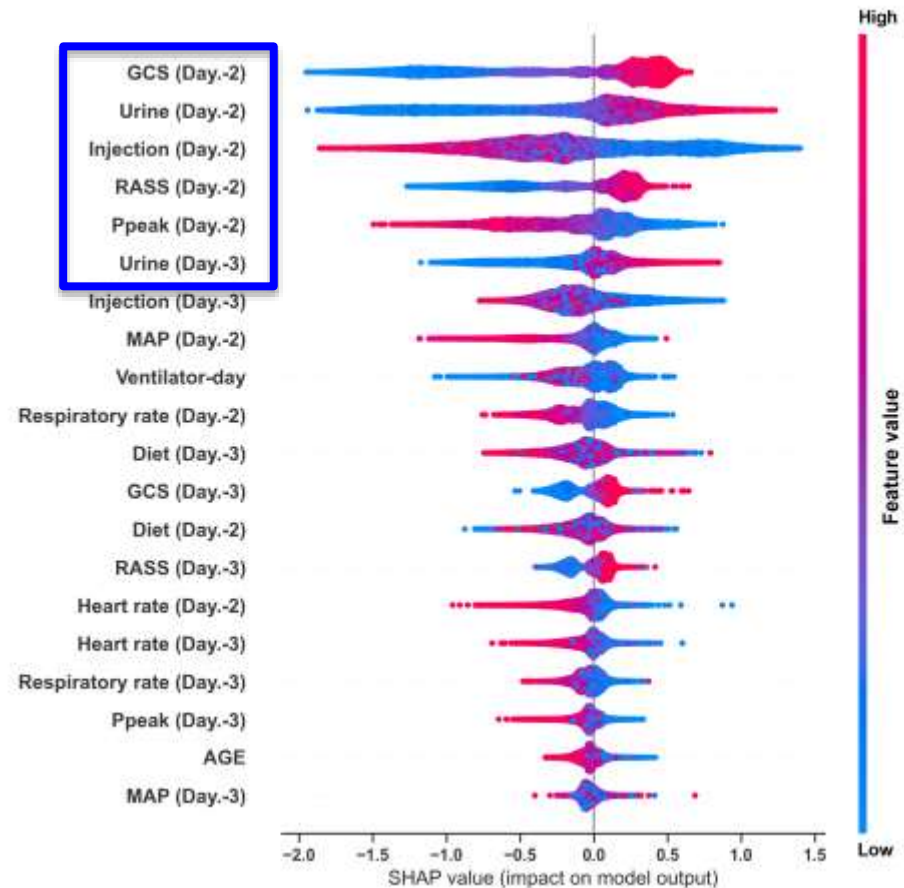
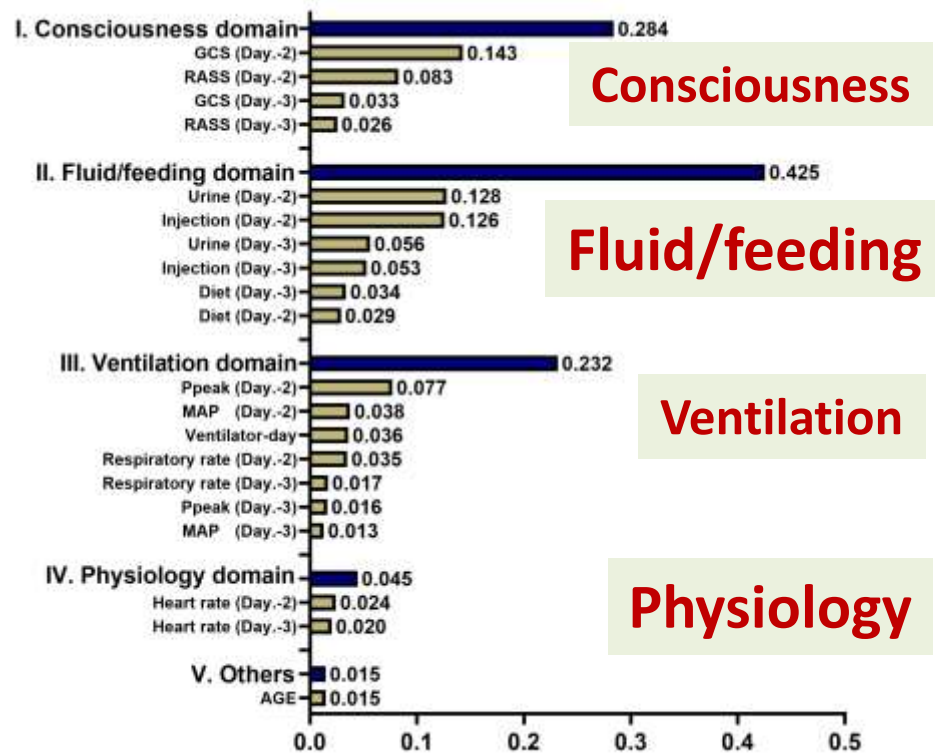
4. Transparency

AI systems and their decisions should be explained in a manner adapted to the stakeholder concerned.

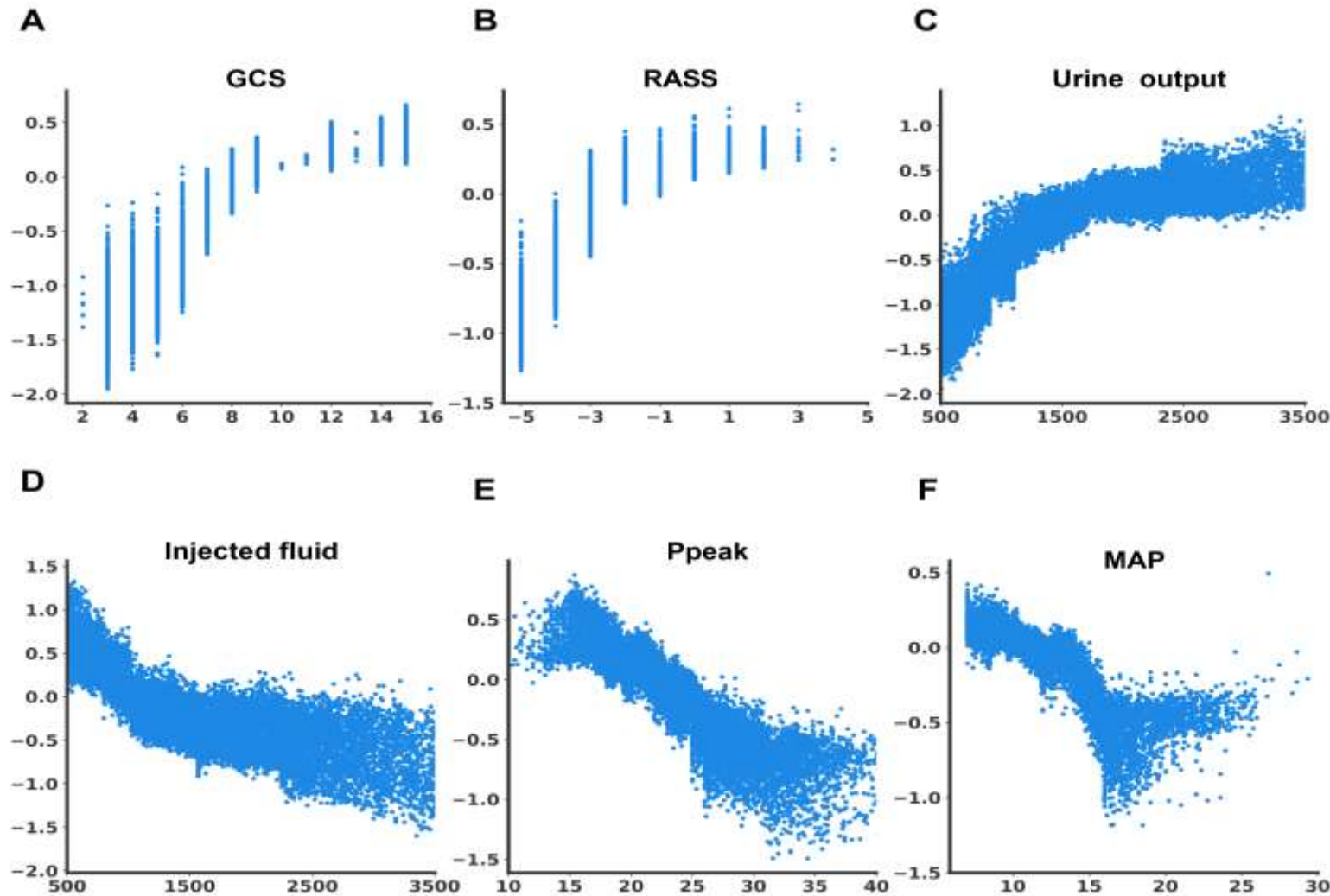


Explainable machine learning approach to predict extubation in critically ill ventilated patients: a retrospective study in central Taiwan

Interpretability of models



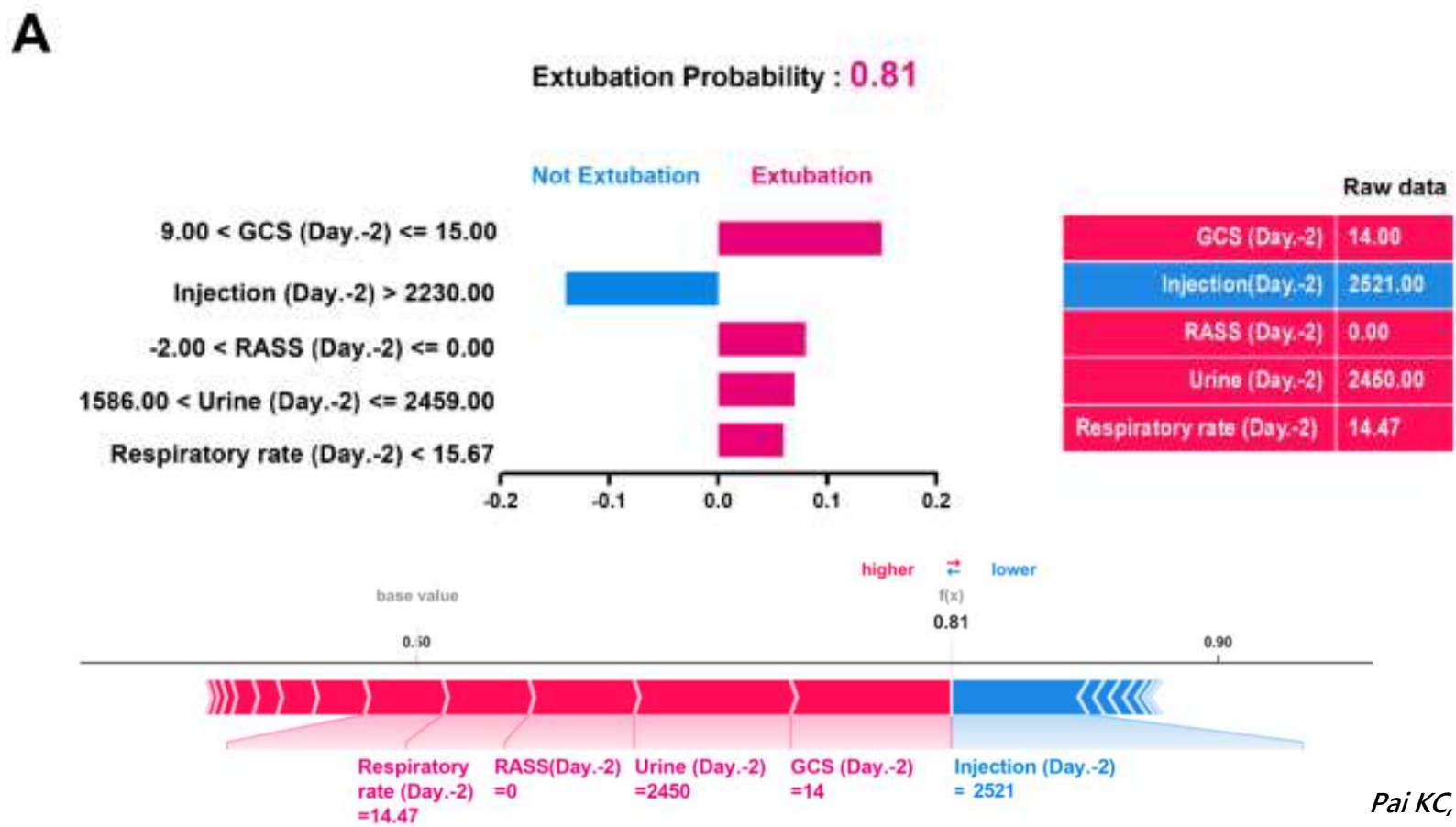
Explainable machine learning approach to **predict extubation** in critically ill ventilated patients: a retrospective study in central Taiwan



Interpretability
→ Feature level

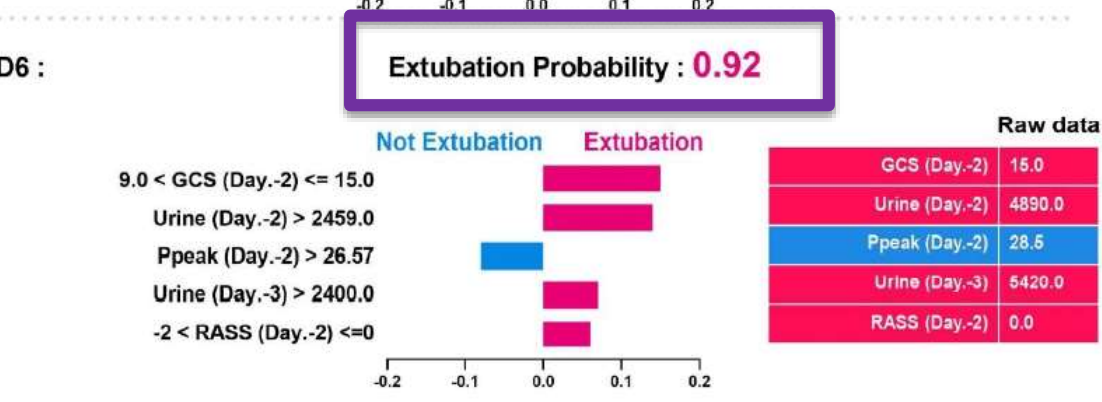
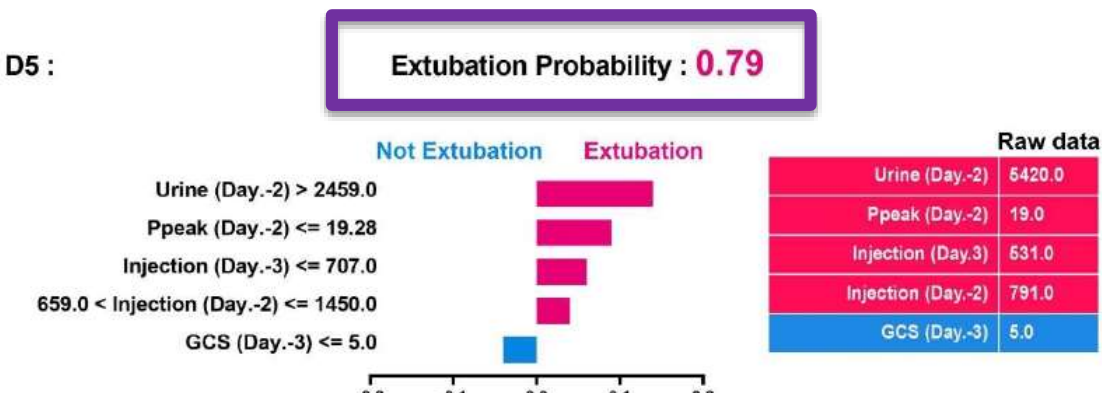
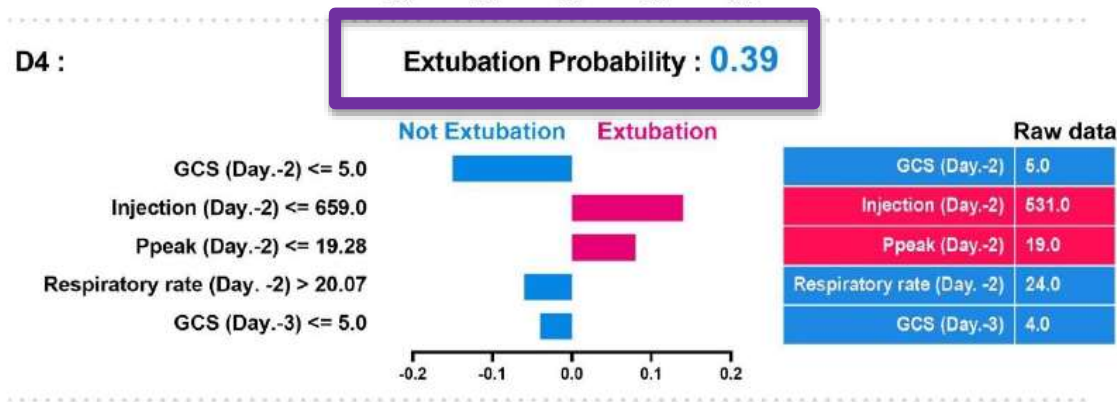
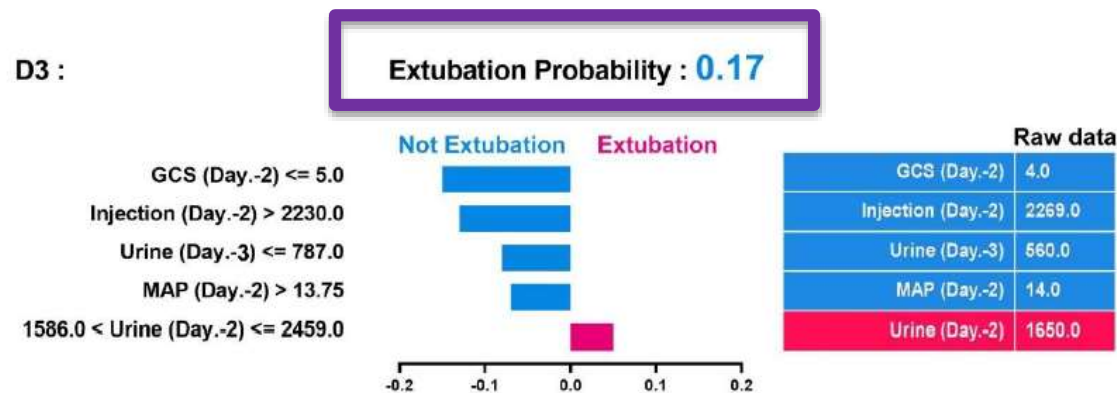
Explainable machine learning to predict long-term mortality in critically ill ventilated patients: a retrospective study in central Taiwan

Interpretability at patient-level



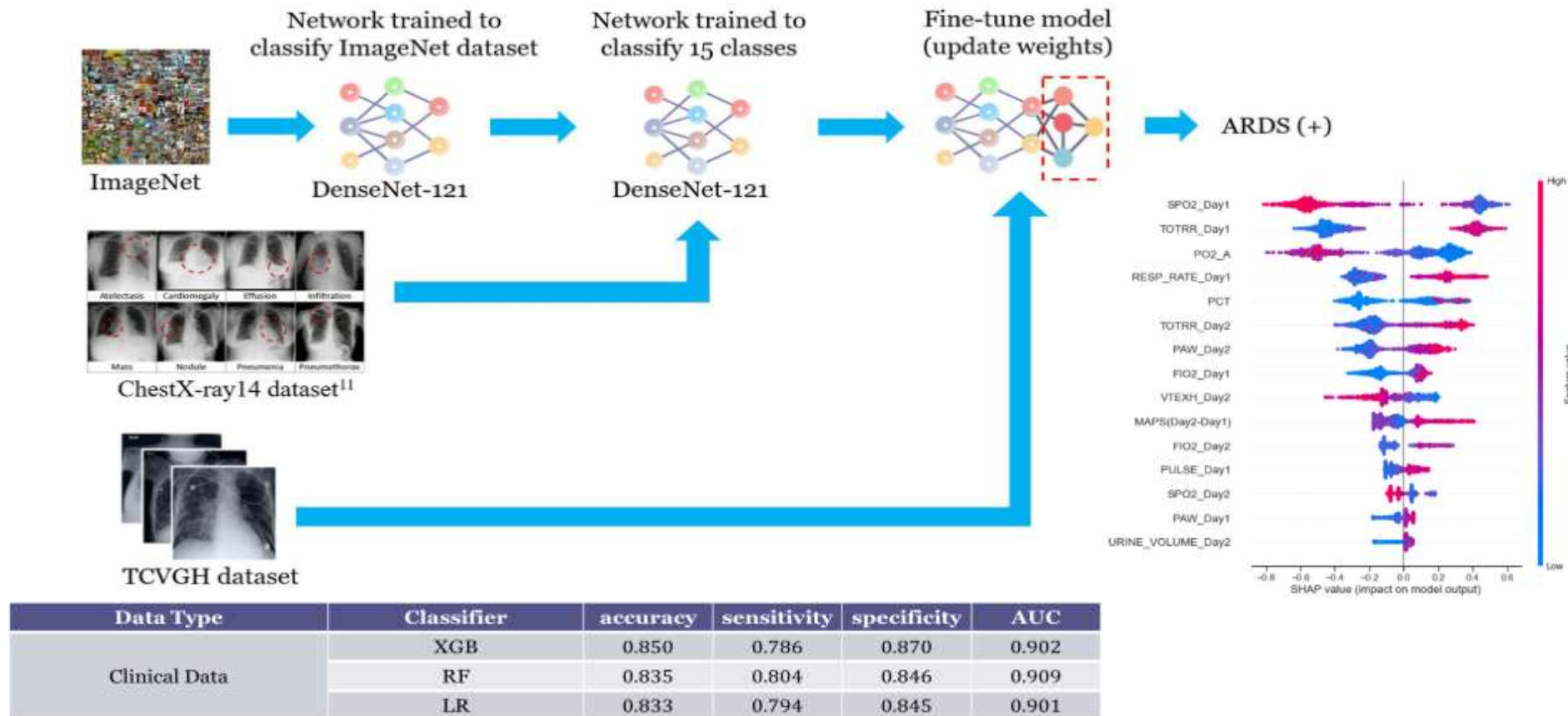
Explainable machine learning approach to predict extubation in critically ill ventilated patients: a retrospective study in central Taiwan

Serial prediction ➔ Automated screen tool



AI-aided ARDS diagnosis

Artificial intelligence–aided diagnosis model **for acute respiratory distress syndrome** combining clinical data and chest radiographs



3P of AI model: Paper-Patent-Product (SaMD)

AKI

> *Health Inf Sci Syst*. 2023 Oct 9;11(1):48. doi: 10.1007/s13755-023-00248-5. eCollection 2023 Dec.

Federated machine learning for predicting acute kidney injury in critically ill patients: a multicenter study in Taiwan

Chun-Te Huang^{1,2}, Tsai-Jung Wang³, Li-Kuo Kuo⁴, Ming-Ju Tsai⁴, Cong-Tat Cia⁵,
Dung-Hung Chiang⁶, Po-Jen Chang⁷, Inn-Wen Chong⁴, Yi-Shan Tsai⁸, Yuan-Chia Chu⁹,
Chia-Jen Liu⁵, Cheng-Hsu Chen¹⁰, Kai-Chih Pai¹¹, Chieh-Liang Wu¹²

Affiliations + expand

PMID: 37822805 PMCID: PMC10562351 DOI: 10.1007/s13755-023-00248-5

ARDS

> *Digit Health*. 2022 Aug 15;8:20552076221120317. doi: 10.1177/20552076221120317. eCollection 2022 Jan-Dec.

Artificial intelligence-aided diagnosis model for acute respiratory distress syndrome combining clinical data and chest radiographs

Kai-Chih Pai¹, Wen-Cheng Chao^{2,3,4}, Yu-Len Huang⁵, Ruey-Kai Sheu⁶, Lun-Chi Chen⁷,
Min-Shian Wang⁶, Shau-Hung Lin⁷, Yu-Yi Yu^{8,9}, Chieh-Liang Wu^{2,3,4}, Ming-Cheng Chan^{3,4}

Affiliations + expand

PMID: 35990108 PMCID: PMC9386858 DOI: 10.1177/20552076221120317

Weaning

> *BMC Anesthesiol*. 2022 Nov 14;22(1):351. doi: 10.1186/s12871-022-01888-y.

Explainable machine learning approach to predict extubation in critically ill ventilated patients: a retrospective study in central Taiwan

Kai-Chih Pai^{1,2}, Shao-An Su², Ming-Cheng Chan^{3,4,5,6}, Chieh-Liang Wu^{5,7,8},
Wen-Cheng Chao^{9,10,11,12}

Affiliations + expand

PMID: 36376785 PMCID: PMC9664699 DOI: 10.1186/s12871-022-01888-y

National Innovation Award 2023



The publications and award represent the **START** of the product.

3P of AI model: Paper-Patent-Product (SaMD)

AKI

ARDS

Weaning



Notice of allowance
May 28th, 2026

Taiwan patent

US patent

Taiwan patent

US patent

Taiwan patent

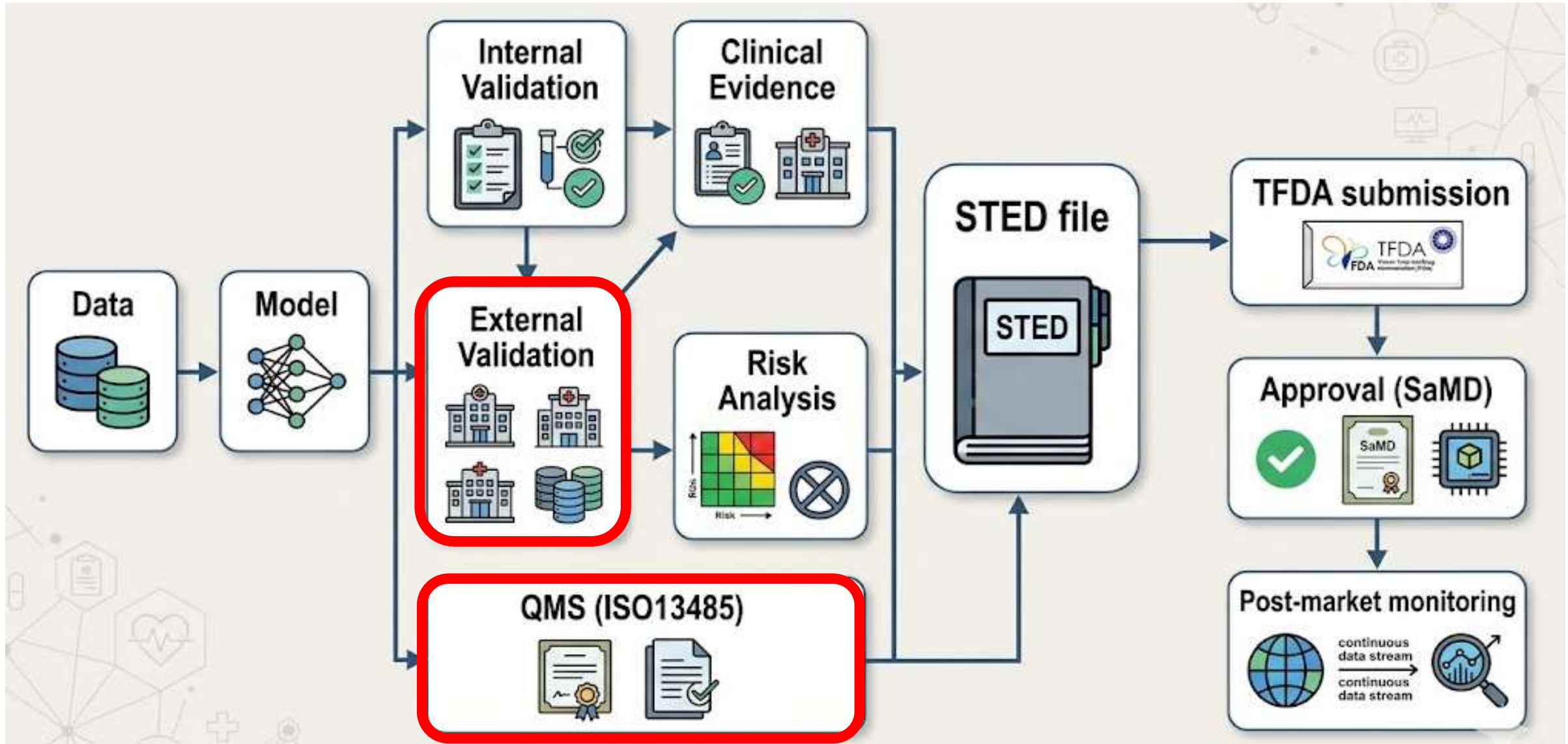
US patent

Patent: 1. Novelty 2. Inventive Step (Non-obviousness) 3. Industrial Applicability (Utility)



- Overseas patent applications and responses for dispute cases.
- Overseas trademark applications and responses for dispute cases.

Step-by-step to acquire certification of SaMD in Taiwan



STED = Summary Technical Documentation

2024th Year MOHW Clinical AI Certification and Validation Center: Taichung Veterans General Hospital Team (VGH Taichung - MacKay - Tung - KMU)

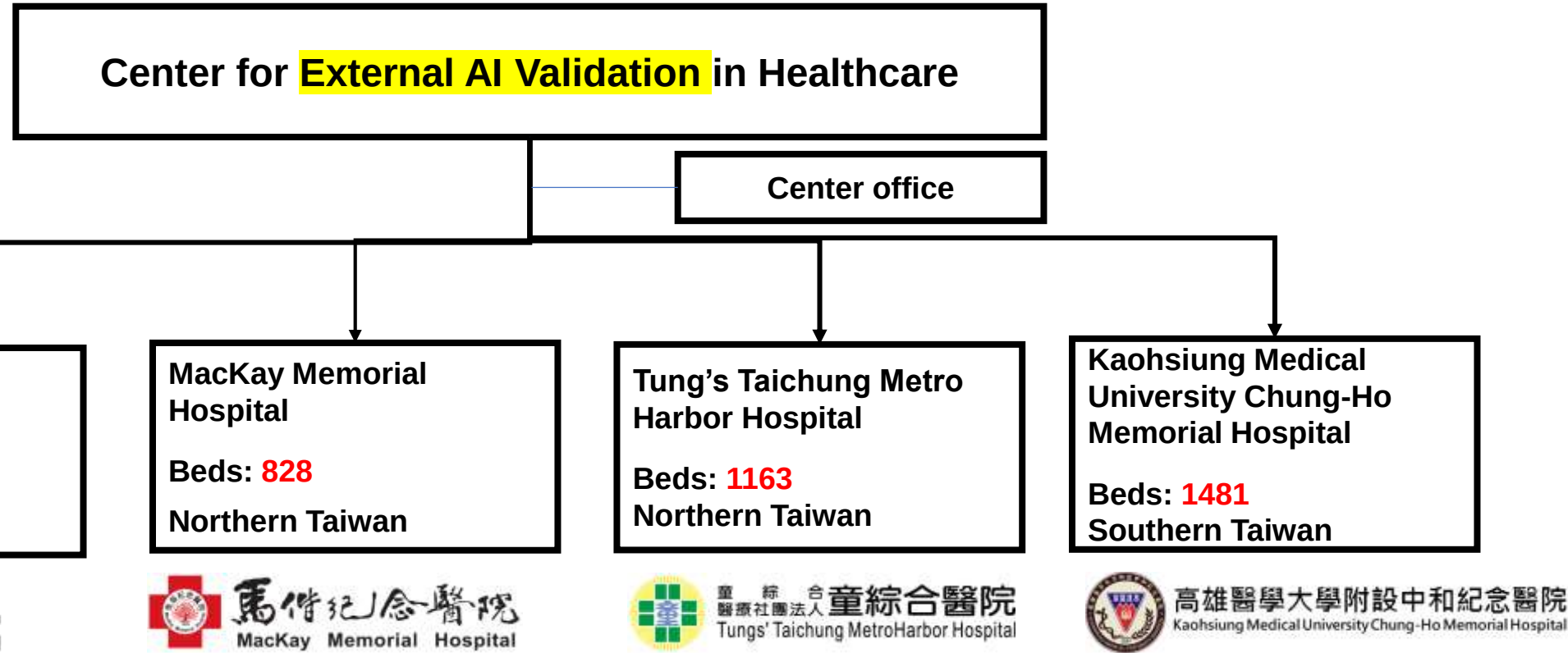
The 4 centers for external AI validation in healthcare

Funded by Ministry of Health and Welfare (MOHW), Taiwan



Center of External AI Validation in Healthcare

TCVGH Team: TCVGH - MacKay - Tung - KMU



TCVGH team including

Project Principal Investigator
Co-Principal Investigator
Project Executive Secretary
Data Scientist

Epidemiologist or Biostatistician
Health Informatics Engineer
AI Engineer
Clinical Researcher (adjusted based on the topic)

Quality Management System (QMS) for SaMD

QMS ensures the SaMD is:

- **Safe**
- **Effective**
- **Traceable**
- **Reproducible**
- **Regulatory-compliant**
- **Be monitored after deployment**

Key International Standards

Domain	Standard / Guidance
Medical Device QMS	ISO 13485
Software Lifecycle	IEC 62304
Risk Management	ISO 14971
SaMD Framework	IMDRF SaMD
AI Governance	ISO/IEC 42001

3P of AI model: Paper-Patent-Product (SaMD)

AKI

ARDS

Weaning



1. QMS (done)
2. External Validation (done)
3. Under TFDA Review (Approved)

Working on
QMS and external
Validation

Real-world application of the AI models in critical care at TCVGH

Dashboard of Department of Critical Care Medicine

Bed condition of the distinct ICUs [Department-level]



Dashboard of Department of Critical Care Medicine

Distinct predicted risks in patients admitted to the RICU [RICU-level]



Dashboard of Department of Critical Care Medicine

Distinct predicted risks in patients admitted to the RICU

[RICU+ Respiratory care-level]



Dashboard of Department of Critical Care Medicine

Distinct predicted risks in **individual** patient admitted to the RICU

[RICU-11—Acute Respiratory Distress Syndrome (ARDS)]



Dashboard of Department of Critical Care Medicine

Distinct predicted risks in **individual** patient admitted to the RICU

[RICU-8—Weaning probability]



Dashboard of Department of Critical Care Medicine

Distinct predicted risks in **individual** patient admitted to the RICU

[RICU-10—Acute Kidney Injury (AKI) probability and explainability]



Real-world ARDS-Dashboard at RICU in TCVGH

1. Prediction + interpretability + interactive dashboard

2. Actionable AI

RICU
Bed allocation



Timer: 15

職稱: 請選取職稱

卡號:

床號: 09

原因:

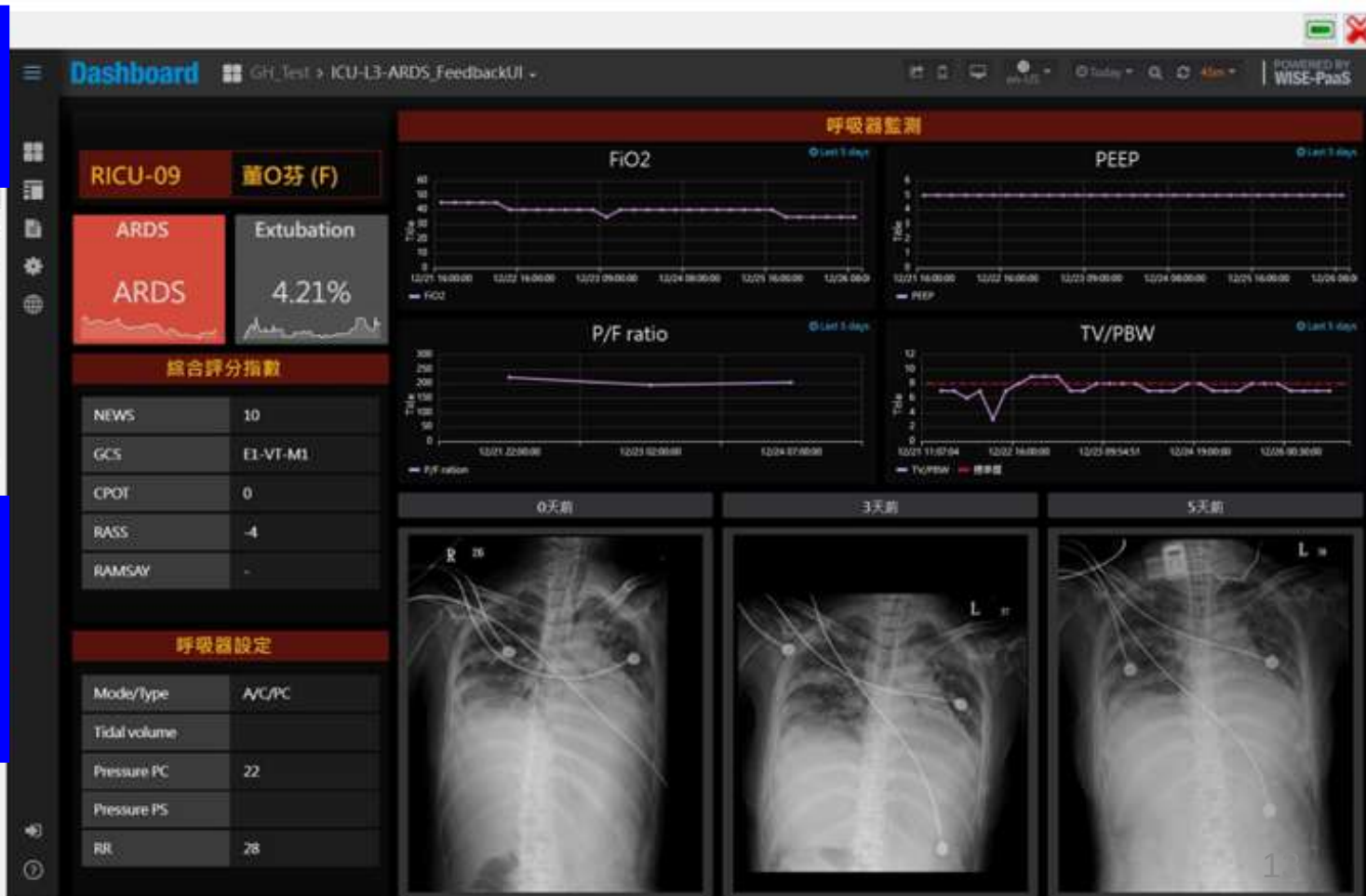
調整註記: ☐ ARDS ☐ Non-ARDS

儲存回饋建議



Actionable AI

1. TV/PBW → <8 cc/kg
2. P/F <150 → Prone
3. NMBAs → Timer
4. Fluid balance
5. Early/trophic feeding



Take Home Message

1. Physicians, as **future users** of medical AI, should be actively involved to design AI-aided healthcare systems.
2. **Explainable** machine learning (XML) is increasingly used in medical AI to enhance trust and willingness to use AI by the clinician.
3. Publications represent the **start** to obtain the patent and the SaMD.
4. **Collaboration** for the external validation and QMS are substantial for SaMD.



臺中榮民總醫院
Taichung Veterans General Hospital



中榮官網



中榮FB

Thank you!